

PHYSICS PRACTICAL SHEETS

Date: 2081/2/1

PN CAMPUS

Class: BSc. 2nd year

Roll No.: 68

Shift: Day

Object of the Experiment (Block Letter)

Experiment No.: 24

Group: P1

Sub.: Physics

Set: 1

OBJECT: TO DETERMINE THE WAVELENGTH OF SODIUM LIGHT USING A PLANE DIFFRACTION GRATING.

APPARATUS REQUIRED:

- | | |
|---------------------|--------------------------|
| (i) A spectrometer | (iv) Diffraction grating |
| (ii) A spirit level | (v) Clamping arrangement |
| (iii) Sodium lamp | (vi) Eye piece |

THEORY:

The process of bending or spreading of light around the corner or edge of the obstacle of aperture is called diffraction. For diffraction the size of the aperture must be comparable to the wavelength of incident waves.

An arrangement which consists of large number of parallel slits having equal width and separated from one another by equal opaque spaces is called diffraction grating.

When a parallel beam of monochromatic light is incident normally on a grating, the transmitted light gives rise to primary maxima in certain directions given by the relation,

$$(a+b) \sin \theta_n = n\lambda$$

where 'a' is the width of a transparency,

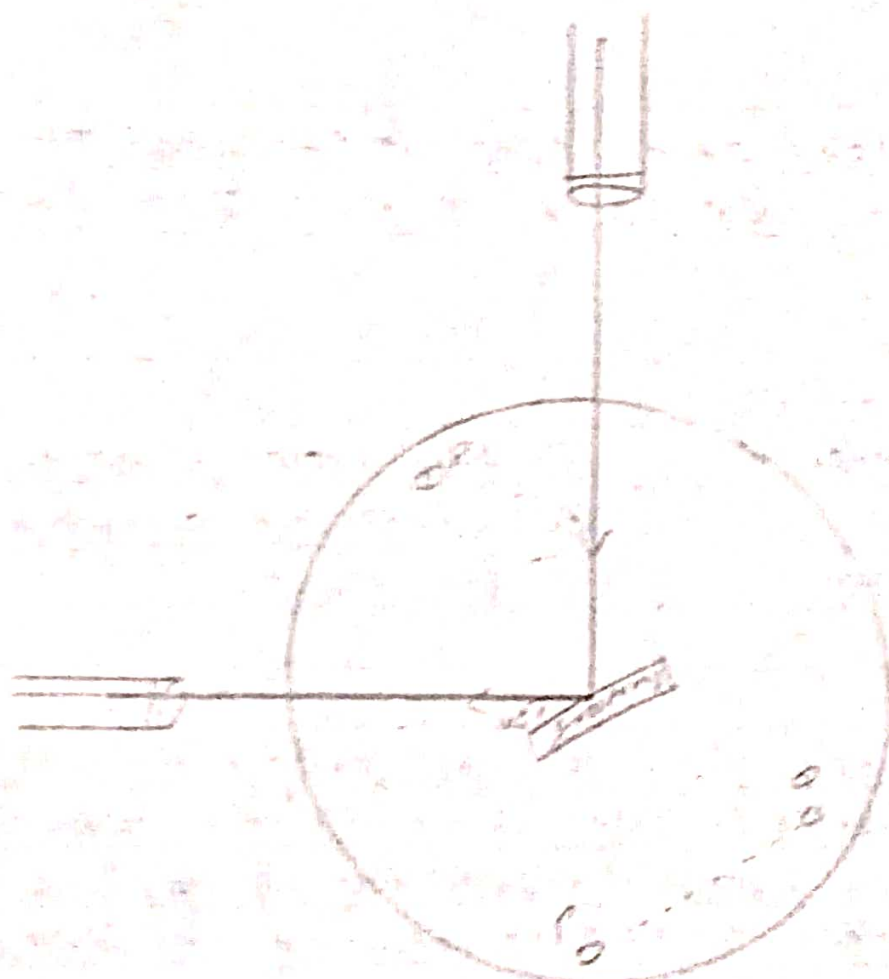


Fig. Fraunhofer diffraction

b that of an opacity, θ the angle of diffraction for the n th order maximum and λ the wavelength of light.

OBSERVATION:

Number of lines per inch on the grating $N = 15000$
 Grating element $(a+b) = \underline{2.54 \text{ cm}}$

In our practical, $(a+b) = \frac{1}{500} \text{ mm} \times 15000 = \frac{1}{500} \times 10^{-3} \text{ m}$

20 division on M.S. $= 10^\circ$

1 division of M.S. $= \frac{1}{2}^\circ$

30 division on V.S. coincides with 29 division on M.S.

1 division of V.S. coincides with 29 division on M.S.

$$\therefore VC = \left(1 - \frac{29}{30}\right) \times \frac{1}{2}^\circ$$

$$VC = \left(\frac{1}{60}\right)^\circ$$

OBSERVATION TABLE:

S.N. Order of Spectral Line	Telescope Reading										Diffraction (20)	θ	Mean θ
	Window	Direct			Left side			Right side					
		M.S.	V.S.	Total	M.S.	V.S.	Total	M.S.	V.S.	Total			
1st order	V ₁	245.5	6	245.6	262.5	5	262.55	228.5	6	228.6	33.98	16.99	16.9825
1st order	V ₂	65	22	65.36	82	20	82.33	48.5	12	48.7	33.63	16.815	
2nd order	V ₁	245.5	1	245.51	263	2	263.03	228.5	101	228.73	34.3	17.15	17.0325
	V ₂	65	9	65.25	82	1	82.06	48	2	48.033	33.983	16.915	

CALCULATION:

Wavelength of sodium light calculated from 1st order spectrum,

$$\lambda_1 = (a+b) \sin \theta$$
$$= \frac{1}{500} \times 10^{-3} \sin (16.9025)$$

$$= 5814 \times 10^{-10} \text{ m}$$

$$\lambda_1 = 5814 \text{ \AA}$$

Again,

$$\lambda_2 = (a+b) \sin \theta$$
$$= \frac{1}{500} \times 10^{-3} \sin (17.0325)$$

$$= 5858 \times 10^{-10} \text{ m}$$

$$\lambda_2 = 5858 \text{ \AA}$$

$$\text{Mean } \lambda = \frac{\lambda_1 + \lambda_2}{2} = \frac{5814 + 5858}{2} = 5836 \text{ \AA}$$

RESULT:

The wavelength of sodium light was found to be 5836 \AA using plane diffraction grating.

ERROR ANALYSIS:

$$\text{Percentage error} = \left| \frac{\lambda_{\text{standard}} - \lambda_{\text{observed}}}{\lambda_{\text{standard}}} \right| \times 100\%$$

$$= \left| \frac{5893 - 5836}{5893} \right| \times 100\%$$

$$= 0.96\%$$

CONCLUSION:

Hence the wavelength of sodium light using plane diffraction grating is $5836 \text{ \AA}$ with 0.96% error.

PRECAUTIONS:

- (i) The grating should be held from the edges and the ruled surface should not be touched.
- (ii) The telescope should be focused on the brightest image of the slit while seeing the reflected image.
- (iii) The ruled surface should face away from the collimator.
- (iv) The light should fall on the whole of the grating surface.